

COMPUTER SCIENCE PBA 2026 MOST EXPECTED GUESS PAPER (FBISE CLASS 12)

TOPIC 1: FUNCTIONS

Question 1

Write a Python function that calculates the square of a number. Continue taking input until a negative number is entered.

Question 2

Write a Python function that calculates the cube of a number.

Question 3

Write a Python function to find the factorial of a number.

Question 4

Write a Python function that determines whether a number is even or odd.

Question 5

Write a Python function that finds the largest of three numbers.

Question 6

Write a Python function that calculates the area of a circle.

Question 7

Write a Python function that calculates the grade of a student according to marks.

Question 8

Write a Python function that returns the sum of first N natural numbers.

Question 9

Write a Python function that counts vowels in a string.

Question 10

Write a Python function that checks whether a number is prime.

TOPIC 2: FILE HANDLING

Question 1

Write a Python program that opens a file and reads its contents line by line.

Question 2

Write a Python program that reads a file character by character.

Question 3

Write a Python program that counts the total number of lines in a file.

Question 4

Write a Python program that counts words in a file.

Question 5

Write a Python program that writes student names into a file.

Question 6

Write a Python program that appends new data to an existing file.

Question 7

Write a Python program that copies contents of one file into another.

Question 8

Write a Python program that displays only the first five lines of a file.

TOPIC 3: LISTS

Question 1

Create a list of marks and print it.

Question 2

Find the maximum and minimum value from a list.

Question 3

Calculate average of values stored in a list.

Question 4

Count the numbers greater than 50 in a list.

Question 5

Print all even numbers from a list.

Question 6

Print all odd numbers from a list.

Question 7

Find the sum of all elements of a list.

Question 8

Search a given number from a list.

Question 9

Sort a list in ascending order.

Question 10

Sort a list in descending order.

TOPIC 4: TURTLE GRAPHICS

Question 1

Draw a square using Turtle Graphics.

Question 2

Draw a rectangle using Turtle Graphics.

Question 3

Draw an equilateral triangle using Turtle Graphics.

Question 4

Draw a pentagon using Turtle Graphics.

Question 5

Draw a hexagon using Turtle Graphics.

Question 6

Draw a star using Turtle Graphics.

Question 7

Draw concentric circles using Turtle Graphics.

Question 8

Draw a house using Turtle Graphics.

TOPIC 5: LIBRARIES

Question 1

Write a Python program using math library to find square root.

Question 2

Write a Python program using math library to calculate power.

Question 3

Write a Python program using random library to generate a random number.

Question 4

Write a Python program using random library to simulate a dice roll.

Question 5

Write a Python program using random library to generate five random numbers.

TOPIC 6: ENTREPRENEURSHIP / MVP

Question 1

Design an MVP for an Online Book Store.

Question 2

Design an MVP for an Online T-Shirt Store.

Question 3

Design an MVP for a Food Delivery App.

Question 4

Design an MVP for an Online Notes Sharing Platform.

Question 5

Design an MVP for a Ride Sharing Service.

Question 6

Identify Frontend and Backend technologies for an Online Shopping Website.

Question 7

Suggest three future improvements for a Student Management System.

Question 8

Evaluate a prototype and suggest improvements.

TOPIC 7: ALGORITHMS & PSEUDOCODES

Question 1

Write an algorithm to find the largest of three numbers.

Question 2

Write an algorithm to find the smallest of three numbers.

Question 3

Write an algorithm to print even numbers from 1 to 100.

Question 4

Write an algorithm to print odd numbers from 1 to 100.

Question 5

Write an algorithm to calculate factorial of a number.

Question 6

Write an algorithm to print multiplication table of a number.

Question 7

Write an algorithm to find sum of first N natural numbers.

Question 8

Write an algorithm to determine whether a number is prime.

Question 9

Write pseudocode for grade calculation.

Question 10

Write pseudocode to calculate average marks of students.

TOPIC 8: SORTING

Question 1

Apply Bubble Sort on:

45, 12, 67, 23, 89

Question 2

Apply Bubble Sort on:

78, 25, 13, 90, 44

Question 3

Apply Insertion Sort on:

20, 15, 30, 10, 5

Question 4

Apply Insertion Sort on:

50, 40, 70, 10, 60

TOPIC 9: SEARCHING

Question 1

Apply Linear Search to find 25.

Question 2

Apply Linear Search to find 80.

Question 3

Apply Binary Search to find 36.

Question 4

Apply Binary Search to find 75.

Question 5

Apply Binary Search to find 50.

TOPIC 10: TRACE TABLES

Question 1

Draw trace table for factorial algorithm.

Question 2

Draw trace table for multiplication using repeated addition.

Question 3

Draw trace table for sum of first 10 numbers.

Question 4

Draw trace table for counting even numbers.

Question 5

Draw trace table for largest number algorithm.

TOPIC 11: LOGIC GATES & TRUTH TABLES

Question 1

Create truth table for $A+B$.

Question 2

Create truth table for $A.B$.

Question 3

Create truth table for $A \oplus B$.

Question 4

Create truth table for $(A+B)C$.

Question 5

Create truth table for $A'B + AB'$.

Question 6

Draw logic circuit for $A+B$.

Question 7

Draw logic circuit for $A.B$.

Question 8

Draw logic circuit for $A'B + AB'$.

TOPIC 12: K-MAPS

Question 1

Simplify:

$$F(A,B) = \Sigma(0,1,2)$$

Question 2

Simplify:

$$F(A,B)=\Sigma(1,2,3)$$

Question 3

Simplify:

$$F(A,B,C)=\Sigma(0,1,2,3)$$

Question 4

Simplify:

$$F(A,B,C)=\Sigma(1,3,5,7)$$

Question 5

Simplify:

$$F(A,B,C)=\Sigma(0,2,4,6)$$

TOPIC 13: DATA VISUALIZATION

Question 1

Create a Bar Chart for monthly sales.

Question 2

Create a Pie Chart for favourite subjects.

Question 3

Create a Line Graph for rainfall data.

Question 4

Create a Scatter Plot for height and weight data.

Question 5

Create a Box Plot for examination marks.

TOPIC 14: DATA COLLECTION

Question 1

Design four interview questions regarding online learning.

Question 2

Design four survey questions regarding reading habits.

Question 3

Design four survey questions regarding social media usage.

Question 4

Design four interview questions regarding student stress.

Question 5

Design a strategy to collect data using interviews.

Question 6

Design a strategy to collect data using surveys.

Question 7

Design a strategy to collect data using prototypes.

Question 8

Design a strategy to collect data using simulations.

Further Important Questions

Write a Python program that takes a string from the user and converts all lowercase letters into uppercase letters. Then display the converted string.

Question:

Write a Python program that converts all uppercase letters of a string into lowercase letters.

Question:

Write a Python program that reverses a string.

Question

Write a Python program that prints the multiplication table of a number entered by the user. After printing the table, the program should again ask for another number and continue until a negative number is entered.